(See Figure 8.2.1.) It is clear that each of the functions f_n is continuous on $[0, 1]$; hence it is Riemann integrable. Either by means of a direct calculation, or by referring to the significance of the integral as an area, we obtain

$$\int_0^1 f_n(x)dx = 1 \quad \text{for} \quad n \geq 2.$$

The reader may show that $f_n(x) \to 0$ for all $x \in [0, 1]$; hence the limit function f vanishes identically and is continuous (and hence integrable), and $\int_0^1 f(x)dx = 0$. Therefore we have the uncomfortable situation that:

$$\int_0^1 f(x)dx = 0 \neq 1 = \lim \int_0^1 f_n(x)dx.$$

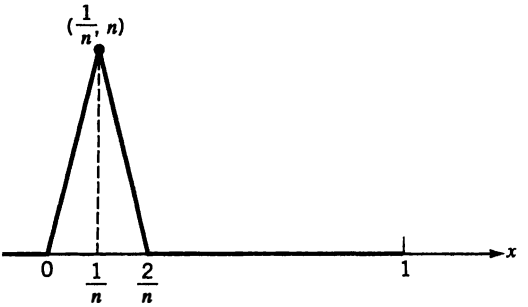


Figure 8.2.1 Example 8.2.1(c)

(d) Those who consider the functions f_n in part (c) to be "artificial" may prefer to consider the sequence (h_n) defined by $h_n(x) := 2nxe^{-nx^2}$ for $x \in [0, 1]$, $n \in \mathbb{N}$. Since $h_n = H'_n$, where $H_n(x) := -e^{-nx^2}$, the Fundamental Theorem 7.3.1 gives

$$\int_0^1 h_n(x)dx = H_n(1) - H_n(0) = 1 - e^{-n}.$$

It is an exercise to show that $h(x) := \lim(h_n(x)) = 0$ for all $x \in [0, 1]$; hence

$$\int_0^1 h(x)dx \neq \lim \int_0^1 h_n(x)dx. \qquad \square$$

Although the extent of the discontinuity of the limit function in Example 8.2.1 (a) is not very great, it is evident that more complicated examples can be constructed that will produce more extensive discontinuity. In any case, we must abandon the hope that the limit of a convergent sequence of continuous [respectively, differentiable, integrable] functions will be continuous [respectively, differentiable, integrable].

It will now be seen that the additional hypothesis of uniform convergence is sufficient to guarantee that the limit of a sequence of continuous functions is continuous. Similar results will also be established for sequences of differentiable and integrable functions.

Interchange of Limit and Continuity

8.2.2 Theorem *Let (f_n) be a sequence of continuous functions on a set $A \subseteq \mathbb{R}$ and suppose that (f_n) converges uniformly on A to a function $f : A \to \mathbb{R}$. Then f is continuous on A.*

Proof. By hypothesis, given $\varepsilon > 0$ there exists a natural number $H := H(\frac{1}{3}\varepsilon)$ such that if $n \geq H$ then $|f_n(x) - f(x)| < \frac{1}{3}\varepsilon$ for all $x \in A$. Let $c \in A$ be arbitrary; we will show that f is continuous at c. By the Triangle Inequality we have

$$\begin{aligned} |f(x) - f(c)| &\leq |f(x) - f_H(x)| + |f_H(x) - f_H(c)| + |f_H(c) - f(c)| \\ &\leq \tfrac{1}{3}\varepsilon + |f_H(x) - f_H(c)| + \tfrac{1}{3}\varepsilon. \end{aligned}$$

Since f_H is continuous at c, there exists a number $\delta := \delta(\frac{1}{3}\varepsilon, c, f_H) > 0$ such that if $|x - c| < \delta$ and $x \in A$, then $|f_H(x) - f_H(c)| < \frac{1}{3}\varepsilon$. Therefore, if $|x - c| < \delta$ and $x \in A$, then we have $|f(x) - f(c)| < \varepsilon$. Since $\varepsilon > 0$ is arbitrary, this establishes the continuity of f at the arbitrary point $c \in A$. (See Figure 8.2.2.) Q.E.D.

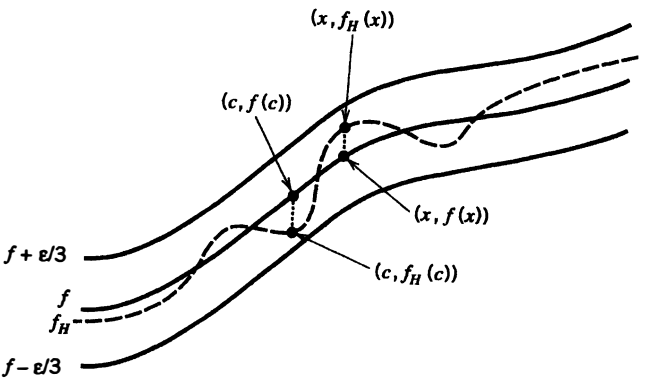


Figure 8.2.2 $|f(x) - f(c)| < \varepsilon$

Remark Although the uniform convergence of the sequence of continuous functions is sufficient to guarantee the continuity of the limit function, it is *not* necessary. (See Exercise 2.)

Interchange of Limit and Derivative

We mentioned in Section 6.1 that Weierstrass showed that the function defined by the series

$$f(x) := \sum_{k=0}^{\infty} 2^{-k}\cos(3^k x)$$

is continuous at every point but does not have a derivative at any point in $\mathbb{R}$. By considering the partial sums of this series, we obtain a sequence of functions (f_n) that possess a derivative at every point and are uniformly convergent to f. Thus, even though the sequence of differentiable functions (f_n) is uniformly convergent, it does not follow that the limit function is differentiable. (See Exercises 9 and 10.)

We now show that if the *sequence of derivatives* (f_n') is uniformly convergent, then all is well. If one adds the hypothesis that the derivatives are continuous, then it is possible to give a short proof, based on the integral. (See Exercise 11.) However, if the derivatives are not assumed to be continuous, a somewhat more delicate argument is required.

8.2.3 Theorem *Let $J \subseteq \mathbb{R}$ be a bounded interval and let (f_n) be a sequence of functions on J to $\mathbb{R}$. Suppose that there exists $x_0 \in J$ such that $(f_n(x_0))$ converges, and that the sequence (f_n') of derivatives exists on J and converges uniformly on J to a function g.*

Then the sequence (f_n) converges uniformly on J to a function f that has a derivative at every point of J and $f' = g$.

Proof. Let $a < b$ be the endpoints of J and let $x \in J$ be arbitrary. If $m, n \in \mathbb{N}$, we apply the Mean Value Theorem 6.2.4 to the difference $f_m - f_n$ on the interval with endpoints x_0, x. We conclude that there exists a point y (depending on m, n) such that

$$f_m(x) - f_n(x) = f_m(x_0) - f_n(x_0) + (x - x_0)\{f_m'(y) - f_n'(y)\}.$$

Hence we have

$$||f_m - f_n||_J \leq |f_m(x_0) - f_n(x_0)| + (b - a)||f_m' - f_n'||_J. \tag{1}$$

From Theorem 8.1.10, it follows from (1) and the hypotheses that $(f_n(x_0))$ is convergent and that (f_n') is uniformly convergent on J, that (f_n) is uniformly convergent on J. We denote the limit of the sequence (f_n) by f. Since the f_n are all continuous and the convergence is uniform, it follows from Theorem 8.2.2 that f is continuous on J.

To establish the existence of the derivative of f at a point $c \in J$, we apply the Mean Value Theorem 6.2.4 to $f_m - f_n$ on an interval with end points c, x. We conclude that there exists a point z (depending on m, n) such that

$$\{f_m(x) - f_n(x)\} - \{f_m(c) - f_n(c)\} = (x - c)\{f_m'(z) - f_n'(z)\}.$$

Hence, if $x \neq c$, we have

$$\left|\frac{f_m(x) - f_m(c)}{x - c} - \frac{f_n(x) - f_n(c)}{x - c}\right| \leq ||f_m' - f_n'||_J.$$

Since (f_n') converges uniformly on J, if $\varepsilon > 0$ is given there exists $H(\varepsilon)$ such that if $m, n \geq H(\varepsilon)$ and $x \neq c$, then

$$\left|\frac{f_m(x) - f_m(c)}{x - c} - \frac{f_n(x) - f_n(c)}{x - c}\right| \leq \varepsilon. \tag{2}$$

If we take the limit in (2) with respect to m and use Theorem 3.2.6, we have

$$\left|\frac{f(x) - f(c)}{x - c} - \frac{f_n(x) - f_n(c)}{x - c}\right| \leq \varepsilon.$$

provided that $x \neq c$, $n \geq H(\varepsilon)$. Since $g(c) = \lim(f_n'(c))$, there exists $N(\varepsilon)$ such that if $n \geq N(\varepsilon)$, then $|f_n'(c) - g(c)| < \varepsilon$. Now let $K := \sup\{H(\varepsilon), N(\varepsilon)\}$. Since $f_K'(c)$ exists, there exists $\delta_K(\varepsilon) > 0$ such that if $0 < |x - c| < \delta_K(\varepsilon)$, then

$$\left|\frac{f_K(x) - f_K(c)}{x - c} - f_K'(c)\right| < \varepsilon.$$

Combining these inequalities, we conclude that if $0 < |x - c| < \delta_K(\varepsilon)$, then

$$\left|\frac{f(x) - f(c)}{x - c} - g(c)\right| < 3\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, this shows that $f'(c)$ exists and equals $g(c)$. Since $c \in J$ is arbitrary, we conclude that $f' = g$ on J. Q.E.D.

Interchange of Limit and Integral

We have seen in Example 8.2.1(c) that if (f_n) is a sequence $\mathcal{R}[a, b]$ that converges on $[a, b]$ to a function f in $\mathcal{R}[a, b]$, then it need not happen that

$$\int_a^b f = \lim_{n \to \infty} \int_a^b f_n. \tag{3}$$

We will now show that *uniform convergence* of the sequence is sufficient to guarantee that this equality holds.

8.2.4 Theorem *Let (f_n) be a sequence of functions in $\mathcal{R}[a,b]$ and suppose that (f_n) converges* ***uniformly*** *on $[a, b]$ to f. Then $f \in \mathcal{R}[a,b]$ and (3) holds.*

Proof. It follows from the Cauchy Criterion 8.1.10 that given $\varepsilon > 0$ there exists $H(\varepsilon)$ such that if $m > n \geq H(\varepsilon)$ then

$$-\varepsilon \leq f_m(x) - f_n(x) \leq \varepsilon \quad \text{for} \quad x \in [a,b].$$

Theorem 7.1.5 implies that

$$-\varepsilon(b-a) \leq \int_a^b f_m - \int_a^b f_n \leq \varepsilon(b-a).$$

Since $\varepsilon > 0$ is arbitrary, the sequence $(\int_a^b f_m)$ is a Cauchy sequence in $\mathbb{R}$ and therefore converges to some number, say $A \in \mathbb{R}$.

We now show $f \in \mathcal{R}[a,b]$ with integral A. If $\varepsilon > 0$ is given, let $K(\varepsilon)$ be such that if $m > K(\varepsilon)$, then $|f_m(x) - f(x)| < \varepsilon$ for all $x \in [a,b]$. If $\dot{\mathcal{P}} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is any tagged partition of $[a, b]$ and if $m > K(\varepsilon)$, then

$$\begin{aligned} |S(f_m;\dot{\mathcal{P}}) - S(f;\dot{\mathcal{P}})| &= \left|\sum_{i=1}^n \{f_m(t_i) - f(t_i)\}(x_i - x_{i-1})\right| \\ &\leq \sum_{i=1}^n |f_m(t_i) - f(t_i)|(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^n \varepsilon(x_i - x_{i-1}) = \varepsilon(b-a). \end{aligned}$$

We now choose $r \geq K(\varepsilon)$ such that $|\int_a^b f_r - A| < \varepsilon$ and we let $\delta_{r,\varepsilon} > 0$ be such that $|\int_a^b f_r - S(f_r;\dot{\mathcal{P}})| < \varepsilon$ whenever $||\dot{\mathcal{P}}|| < \delta_{r,\varepsilon}$. Then we have

$$\begin{aligned} |S(f;\dot{\mathcal{P}}) - A| &\leq |S(f;\dot{\mathcal{P}}) - S(f_r;\dot{\mathcal{P}})| + \left|S(f_r;\dot{\mathcal{P}}) - \int_a^b f_r\right| + \left|\int_a^b f_r - A\right| \\ &\leq \varepsilon(b-a) + \varepsilon + \varepsilon = \varepsilon(b-a+2). \end{aligned}$$

But since $\varepsilon > 0$ is arbitrary, it follows that $f \in \mathcal{R}[a,b]$ and $\int_a^b f = A$. Q.E.D.

The hypothesis of uniform convergence is a very stringent one and restricts the utility of this result. In Section 10.4 we will obtain some far-reaching generalizations of Theorem 8.2.4. For the present, we will state a result that does not require the uniformity of the convergence, but does require that the limit function be Riemann integrable. The proof is omitted.

8.2.5 Bounded Convergence Theorem *Let (f_n) be a sequence in $\mathcal{R}[a,b]$ that converges on $[a, b]$ to a function $f \in \mathcal{R}[a,b]$. Suppose also that there exists $B > 0$ such that $|f_n(x)| \leq B$ for all $x \in [a,b]$, $n \in \mathbb{N}$. Then equation (3) holds.*